

ΓΡΑΦΙΚΗ - PROJECT - 28/6/2024

ΣΤΑΥΡΟΠΟΥΛΟΥ ΕΛΕΝΗ - 2019030070

Email επικοινωνίας: elena1stavropoulou@gmail.com

Project Link:

<https://drive.google.com/file/d/1ysG8V3JTkwuDfsewOSmT78h8nlpIlpJx/view?usp=sharing>

Περιγραφή του κώδικα:

MainMenu: Handles main menu functionality, including starting the game and quitting the application.

Timer: This script manages a countdown timer and loads the next scene when time runs out.

Item: This script is a base class for inventory items, providing basic functionality for using and removing items.

TreeTrunk: This script handles the collection of tree trunks and spawns a fire when three trunks are collected.

Mushroom: This script increases the player's health when a mushroom item is used.

Axe: This script allows the player to use an axe item to chop down trees or interact with other objects.

Inventory: Manages the player's inventory, allowing items to be added and removed.

InventoryUI: Updates the inventory UI to reflect changes in the player's inventory. Subscribes to inventory events.

InventorySlot: Represents a slot in the inventory UI. Handles item addition, removal, and slot clearing.

Interactable: Base class for interactable objects in the game. Manages interaction triggers and focus.

ItemPickUp: Allows the player to pick up items in the game world and adds them to the inventory.

TreeChopping: Represents a tree that can be chopped down with an axe. Spawns logs and updates tasks text upon interaction.

HealthBar: Manages the UI representation of a health bar, adjusting its value and orientation relative to the camera.

PlayerStats: Controls player's health, including damage, healing, and handling game over conditions.

PlayerMotor: Facilitates player navigation using NavMeshAgent, including movement towards a target and rotation to face it.

PlayerAnimator: Manages player animations based on movement speed, ensuring smooth transitions between different locomotion states.

PlayerControl: Handles player input for movement and interaction, using raycasting to determine movement destinations and interaction targets.

CameraControl: Controls camera position, rotation, and zoom relative to the player, enabling zoom and rotation based on user input.

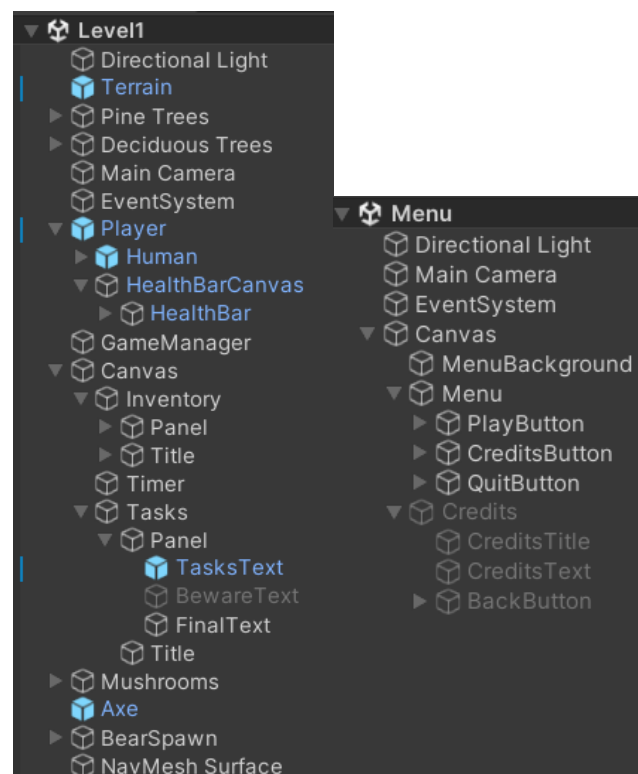
BearSpawn: Spawns bear enemy at a specific time interval from specified spawn points, displaying a warning text upon bear spawn.

BearControl: Controls behavior of bear enemies, including chasing and attacking the player, as well as reacting to nearby fires.

BearMotor: Handles navigation of bear enemies using NavMeshAgent, enabling them to chase the player within a specified range.

BearAnimator: Manages animations for bear enemies based on movement speed and triggers for attacking, chasing, and reacting to fires.

Hierarchy Screenshots:



Level2 and Level3 are similar to Level1 Hierarchy.

Game Design Document (GDD)

Bear Chase: A Test of Survival

Playable Character: The player controls a survivor character in a forest setting.

Goal: The player must survive for a set duration by managing their health, avoiding threats and utilizing resources.

Scenario:

- Health Management: The playable character's health decreases over time and even more when attacked by a bear.
- Survival Tasks:
 - Collect Food: Pick mushrooms to eat and increase health.
 - Find Tools: Locate an axe to cut down trees and gather the tree trunks.
 - Create Fires: Use the tree trunks to start fires that scare away the bear.
- Bear Threat: The player is constantly pursued by a bear that attacks if it gets too close unless deterred by a fire.
- Timer: The game has a timer, set to 3 minutes, and the player must survive until it runs out to win the game.

Hostile NPC (Bear): It is spawned after 1.5 minutes since the game has started at a random position within a specified area and immediately starts chasing the player. If within range, it attacks the player, unless a fire is nearby.

Interactable Objects:

- Mushroom: Collectible food item.
- Axe: Tool found in the environment used to cut down trees.
- Tree: Can be cut down with an axe to obtain tree trunks.
- Tree Trunk: Collectible item from trees used to create fires.

Obstacles:

- Tree: Initially blocks paths and can be cut down for resources.
- Mountains: Static obstacles that block the player's movement.

Boosters:

- Mushroom: Consuming mushrooms increases the player's health.
- Fire: Created using at least three tree trunks to scare away the bear and provide temporary safety.

Controls:

- Movement: Left mouse button to move the character to a point.
- Interact: Right mouse button to move the character to the interactable object and interact with it.
- Inventory: I key to close/open the inventory. Left mouse button on an inventory slot to use its item.

User Interface (UI):

- Health Bar: Displays the player's current health status.
- Inventory: Shows the items the player has collected.
- Timer: Displays the remaining time the player needs to survive.

Assets used from Unity Asset Store:

<https://assetstore.unity.com/packages/3d/characters/humanoids/humans/free-modular-character-fantasy-rpg-human-male-228952>

<https://assetstore.unity.com/packages/3d/characters/animals/free-stylized-bear-rpg-forest-animal-228910>

<https://assetstore.unity.com/packages/3d/props/exterior/upixelator-campfire-3d-pixel-art-277510>

<https://assetstore.unity.com/packages/3d/environments/landscapes/handpainted-forest-pack-lite-176577>

<https://assetstore.unity.com/packages/3d/props/tools/free-stylized-tools-248042>

<https://assetstore.unity.com/packages/3d/vegetation/pbr-mushroom-pack-lite-56263>

<https://assetstore.unity.com/packages/2d/gui/fantasy-wooden-gui-free-103811>

<https://assetstore.unity.com/packages/2d/textures-materials/sky/fantasy-skybox-free-18353>

<https://assetstore.unity.com/packages/3d/environments/landscapes/terrain-sample-asset-pack-145808>

<https://assetstore.unity.com/packages/3d/environments/fantasy-landscape-103573>